CLASS 9 · SCIENCE · NCERT

Date: _____ / _____ / _____

Name: _____

Chapter 5 · Topic 1
The Fundamental Unit of Life

📖 *What is a Cell? + Cell Theory*

🔬 **The Big Idea:** All living things — from the tiniest bacteria to the blue whale — are made of **cells**. The cell is the *basic structural and functional unit of life*. Just like bricks make a building, cells make a living organism.

📌 KEY DEFINITION

DEFINITION

Cell

The **smallest structural and functional unit** of a living organism that is capable of carrying out all basic life processes independently.

Omkar EduHub 9862334721

🕰️ HISTORY OF CELL DISCOVERY

1665

Robert Hooke 🇬🇧

Observed thin slices of **cork** under a self-made microscope. Saw tiny box-like structures and called them **"cells"** (from Latin *cellula* = small room). He saw *dead cells* — only cell walls remained.

1674

Anton van Leeuwenhoek 🇳🇱

First to observe **living cells** — bacteria and protozoa in pond water using an improved microscope. Called them **"animalcules."**

1831

Robert Brown 🇬🇧

Discovered the **nucleus** in plant cells. Established that the nucleus is an important component of the living cell.

1839

Purkinje 🇨🇿

Coined the term **"Protoplasm"** for the living jelly-like substance inside cells.

⚗️ CELL THEORY

Schleiden (1838)

German botanist. Said **all plants** are made up of cells.

Schwann (1839)

German zoologist. Said **all animals** are made up of cells.

Virchow (1855)

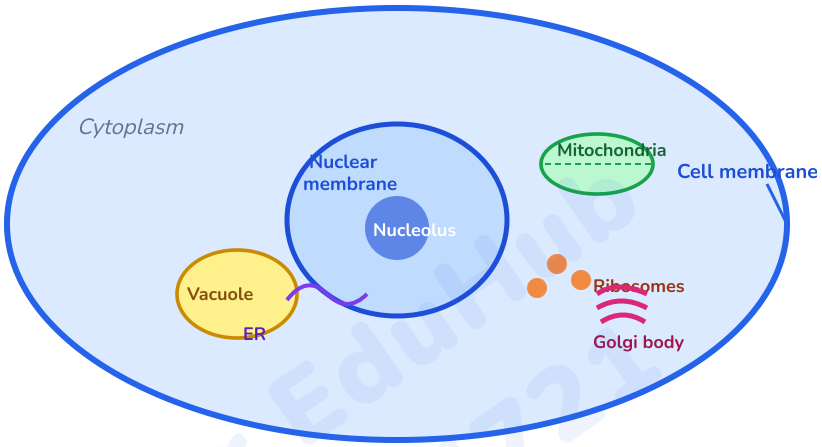
Added: New cells arise from **pre-existing cells** (Omnis cellula e

cellula).

✦ *The 3 Postulates of Cell Theory*

1. **All living organisms** are composed of one or more cells.
2. The **cell is the basic unit** of structure, function, and organization in all living organisms.
3. All cells arise from **pre-existing cells** — cells cannot be created from non-living matter.

🔬 SIMPLE CELL DIAGRAM (ANIMAL CELL)



☝ *Draw and label this diagram in your exam. Learn at least 6 parts.*

Omkar EduHub 9862334721

⚡ QUICK REVISION TABLE

SCIENTIST	YEAR	CONTRIBUTION
Robert Hooke	1665	Discovered dead cells in cork; coined the term "cell"
Leeuwenhoek	1674	Observed first living cells (bacteria, protozoa)
Robert Brown	1831	Discovered the nucleus in plant cells
Schleiden	1838	All plants are made of cells
Schwann	1839	All animals are made of cells
Virchow	1855	New cells arise from pre-existing cells
Purkinje	1839	Coined the term "Protoplasm"

⭐ EXAM TIPS

- Hooke saw **dead cells** — cell walls only. Don't write "living."

⚠️ COMMON MISTAKES

- ❌ Hooke discovered *living* cells → ✅ He saw *dead* cork cells.

- Cell Theory = Schleiden + Schwann + Virchow (all three names).
- "Omnis cellula e cellula" quote = Virchow. Very common 1-mark Q.
- The word "cell" = Latin *cellula* = small room. Good for 1-mark.

- ❌ Leeuwenhoek proposed Cell Theory → ✅ It was Schleiden & Schwann.
- ❌ Schleiden studied animals → ✅ He studied *plants*.
- ❌ Cell = only structural unit → ✅ It is *both* structural AND functional.

🧠 **MEMORY TRICK (MNEMONIC)**

SSV → **S**chleiden (plants) + **S**chwann (animals) + **V**irchow (pre-existing cells)
Remember: "Scholars Study Very carefully" → Schleiden, Schwann, Virchow

Hooke → Hook-shaped cork slices → Dead cells → 1665

🏷️ KEYWORDS TO REMEMBER

`Cell` `Cork` `Cellula` `Protoplasm` `Nucleus` `Cell Theory` `Animalcules` `Pre-existing` `Omnis cellula e cellula` `Structural unit` `Functional unit`

✍️ SELF-TEST (ANSWER WITHOUT LOOKING!)

1. Who coined the term "cell" and in which year? *[_____]*
2. What did Robert Hooke observe under the microscope? *[_____]*
3. Write the Cell Theory in your own words. *[_____]*
4. What does "Omnis cellula e cellula" mean and who said it? *[_____]*
5. What is protoplasm? Who coined this term? *[_____]*

Omkar EduHub 9862334721